

Satvik Sharma

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Technical Skills

Programming & Web Technologies: C++, Python, JavaScript, HTML, CSS, React.js, RESTful APIs

Databases & Cloud platforms: SQL, MongoDB, AWS, Docker, Git, GitHub

AI/ML & MLOps: Scikit-learn, XGBoost, Pandas, Numpy, LangChain, RAG, DVC, MLflow, DagsHub

UI/UX Design & Productivity Tools: UI/UX Design, Figma, Microsoft Office (Excel, PowerPoint)

Experience

FOSSEE Semester Long Internship (IIT Bombay)

Remote

Software Engineer Intern

Feb 2026 – June 2026

- Contributing to the development of a web-based open-source electronics simulation and learning platform.
- Implementing role-based dashboards, classroom management workflows and a gamified progression system.

Excelerate

Remote

Project Management Associate Intern & Team Lead

Oct 2025 – Nov 2025

- Led the team in the strategic design and structuring of four new experiential internship programs, defining problem statements, deliverables, and skill outcomes.
- Managed project execution, delegation, and timelines using a Project Management tool (Trello) to ensure alignment with project goals.

Projects

SolarCarbon AI | *Python, FastAPI, Mendeley Data* | [Deployed Link](#)

- Developed a platform for users to predict their earned carbon credits by using renewable energy sources.
- Utilized the self-authored Mendeley dataset of thermal plant emissions to train a Voting Regressor ensemble (HistGradient-Boosting, RandomForest, ExtraTrees) to predict grid emission factors using physics-based feature engineering.
- Developed a deep preprocessing pipeline with RobustScaler and PolynomialFeatures to handle non-linear relationships, optimizing for high R^2 and low RMSE scores.
- Built a performance-optimized FastAPI backend to calculate real-time carbon credits by integrating live solar irradiance data from the NASA Power API.

OpenHW Studio | *React.js, Node.js, Express.js, MongoDB, Docker* | [Deployed Link](#)

- Engineered a full-stack web platform for open-source electronics simulation and learning under FOSSEE, IIT Bombay, enabling students and educators to design and simulate hardware circuits in the browser.
- Developed role-based dashboards for students, instructors, and admins with distinct permission layers, and built classroom management workflows to streamline course creation and assignment handling.
- Architected a gamified progression system with component learning and level progression after successfully completing the project with a score of 80%.

RAG-Powered AI Code Reviewer | *FastAPI, LangChain, ChromaDB, Sentence Transformers* | [Deployed Link](#)

- Engineered an automated code reviewer deployed as a GitHub Webhook, solving the “institutional memory” gap in tools like GitHub Copilot by autonomously posting context-aware review summaries on new Pull Requests via RAG.
- Built a FastAPI backend to extract and chunk PR differences via the GitHub API, retrieving precedents from a ChromaDB vector store (Sentence Transformers, `all-MiniLM-L6-v2`), and orchestrated a LangChain + Groq (`llama-3.1-8b`) pipeline to post JSON-structured feedback directly into GitHub PR comments.

Achievements

- Hackathon Finalist:** “Automation using Agentic AI” by NASSCOM and Gnani.ai. Built an intelligent AI agent using Inya.ai for Consumer Durables, advancing to the Buildathon Final Review round.
- Carbon Emission Dataset:** Published on Mendeley Data. Performed dataset integration, adding and verifying latitude-longitude columns to support carbon credit research.

Certifications

- Figma UI/UX Design – Udemy (2024)
- Applied ML in Python – University of Michigan, Coursera (2025)
- Postman API – Postman (2024)
- Data Analytics Job Simulation – Deloitte, Forage (2025)
- ML for multimedia forensics – ICTAcademy NIT Patna and Jamshedpur (2025)
- NPTEL Cloud Computing – IIT Kharagpur (2025)
- JLPT N5 - Japanese Language Proficiency Test (2025)

Education

VIT Bhopal University

September 2023 – May 2027

B.Tech Computer Science and Engineering (AIML), CGPA: 8.19/10.0

Shore, MP

Saint Peter Senior Secondary School

Completed: 2023

Class XII (10+2)

Chandigarh

Saint Soldier's Divine Public School

Completed: 2021

Class X

Panchkula, Haryana